

SQL Joins

Primary key - a column in a table that uniquely identifies each row

foreign key - A column in one table that refers to the primary key.

customers		
Primary key	customer id	name
	1	Rafi
	2	Levato

orders → foreign key			
Primary key	order-id	customer-id	amount
	101	1	300
	102	1	150
	103	3	500

Inner Join

Allows us to join two or more tables on a common column then give us the rows that are matching in both tables

Select column1,
column2,
column3

From table 1 AS A

INNER JOIN table 2 AS B

ON A.common-column = B.common-column;

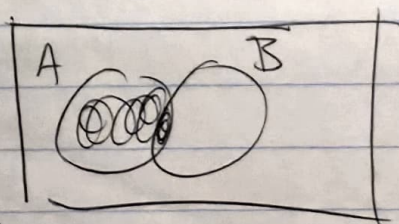
Select customer-id,
name,
amount
from customer AS A
INNER JOIN orders AS B
ON A.customer-id = B.customer-id;

2) select customer-id,
name,
amount

from customers
INNER JOIN orders
ON customer

Left Join

- Returns everything from the left table, and adds matches from the right table
- if no match, right side becomes null



Select column(s) → left table
from table AS A
Left Join table 2 AS B → right table
ON A.common-column = B.common-column.

customers		orders		
customer-id	name	order-id	customer-id	amount
1	Rofi	101	1	300
2	Herato	102	1	150
		103	3	500

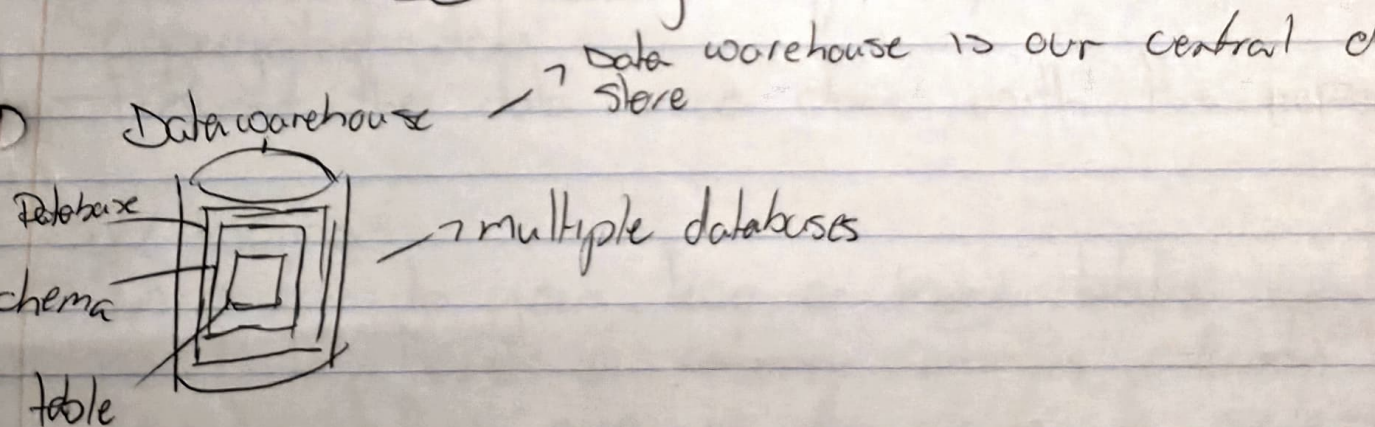
select customer-id,
name,
amount

from customers AS A
left join orders AS B

ON A.customer-id = B.customer-id;

customer-id	name	amount
1	Rofi	300
1	Rofi	150
2	Herato	Null

SQL Spambok pt.1



Get catalog on databricks is the datawarehouse
workspace is the database
default is the schema

- In one datawarehouse you can have multiple databases
- In one database you can have multiple schemas
 - In one schema you can have multiple tables

- Data is stored in tables and its called structured data (rows & columns)
- We use SQL to extract insights from our data

key information that help us make decisions based on data

Big data is a large volume of data
You take the big data and apply SQL to aggregate your data (summarize data)

- You build a powerbi report from the aggregated data
- Present the data once done with the report

SQL Joins

- In order to join two or more tables you need to have a ~~column~~ common column
- Joining tables gives us the ability to select any column between the tables that have been joined together

Inner Join

- It only gives out what is common between the tables
- Returns the rows that are matching in both tables

Students

Course

Students-id	name
1	Ayanda
2	Kabelo
3	Zanele

course id	Student-id	course
201	1	Maths
202	3	Science
203	4	English

Select column(s)

From table 1 AS A

INNER Join table 2 AS B

ON A. common-column = B. common-column

Select student-id,

name,

course

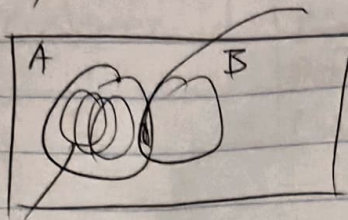
From Students AS A

Inner Join course AS B

ON A. student-id = B. student-id;

Students id	name	course
1	Ayanda	maths
3	Zanele	Science

Left Join



I want only the rows that match the other table

I want every row in the left table, base table

Select column(s) $\rightarrow$ you can choose any column from the tables you are joining

from table 1 AS A

Left Join table 2 AS B $\leftarrow$ right table

ON A.common-column = B.common-column;

Sales		orders		
customer-id	Name	order-id	customer-id	amount
1	Rofi	101	1	300
2	Muuto	102	2	150
		103	3	500

Write a sql script to get the ~~cost~~ customer-id, name, amount

```
SELECT customer-id,  
       name,  
       amount
```

```
FROM customers AS A
```

```
LEFT JOIN orders AS B
```

```
ON A.customer-id = B.customer-id;
```

Your outcome will contain all the rows in the left table

customer_id	name	amount
1	Rofi	300
2	Muvko	150

Departments		Employees		
dept-id	dept-name	emp id	emp-name	dept-id
10	Sales	1	Nee	10
20	IT	2	Rofi	20
30	HR	3	Thabo	20
40	Legal	4	Lesedi	50

I want the dept-id, dept-name, emp-name

```
select emp-id,
       dept-id,
       dept-name,
       emp-name
```

from employees AS A

Left Join departments AS B

ON A.dept-id = B.dept-id;

emp-id	emp-name	dept-id	dept-name
1	Nee	10	Sales
2	Rofi	20	IT
3	Thabo	20	IT
4	Lesedi	50	NULL

Students

Students-id	name
1	Sam
2	Lulu
3	Ben

Clubs

Club-id	Student-id	club-name
10	1	Chess
11	2	Debate
12	4	Soccer

I want to know all the students that play sports, and what sport they play

Select student-id,
name,
club-name

From Students AS A

Left Join Clubs AS B

ON A.Student-id = B.Student-id;

Students-id	Name	club-name
1	Sam	Chess
2	Lulu	Debate
3	Ben	Null

Employees

Employee ID	FirstName	LastName	Department	Salary	City	Age
1	John	Doe	IT	70000	New York	35
2	Alice	Smith	HR	60000	Los Angeles	40
3	Bob	Johnson	Finance	75000	Toronto	45
4	David	Brown	IT	72000	London	30
5	Emma	Wilson	Sales	65000	Sydney	28
6	Michael	Clark	Finance	80000	New York	50

Projects

Project ID	Employee ID	Project Name	Budget	Start Date	Status
101	1	AI Development	100000	2024-01-10	Completed
102	2	Employee Training	50000	2024-02-15	Ongoing
103	1	Cybersecurity Audit	75000	2024-03-05	Pending
104	3	Financial Analysis	90000	2024-04-12	Ongoing
105	5	Market Expansion	65000	2024-05-20	Completed
106	6	Risk Management	80000	2024-06-10	Pending

select EmployeeID,
 FirstName, ProjectID, FirstName,
 LastName, LastName,
 Department, Department,
 Salary, Salary,
 ProjectID,
 ProjectName,
 Budget,
 Start Date

from Employees AS A
Inner Join Projects AS B
ON A.EmployeeID = B.EmployeeID;

Left JOIN

Select EmployeeID,
FirstName,
LastName,
Department,
Salary,
ProjectID,
ProjectName,
Budget,
Status

from EmployeeID AS A
Left Join Projects AS B
ON A.EmployeeID = B.EmployeeID;

Outer Join

Select EmployeeID,
FirstName,
LastName,
Department,
Salary,
ProjectID,
ProjectName,
Budget
and status

From Employees AS A

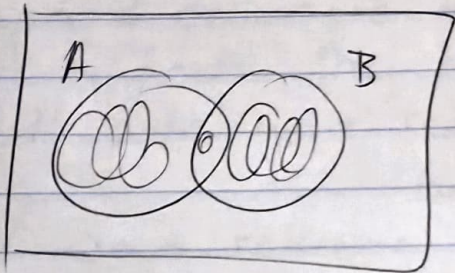
the full outer join Projects AS B

ON B.EmployeeID = A.EmployeeID;

NULL - empty space
- unknown value
- missing value
- blank

Full Outer Join

- Returns all the rows from both the tables



- Because it is a join, you can select any column from tables you are
- A full outer join returns everything from both tables
- if a row matches (same key on both sides)
 - it combines them into one row
- if a row exists ~~only~~ only in the left table $\rightarrow$ it still shows up, with NULL for the right table columns
- if a row exists only in the ~~left~~ ^{right} table it still shows up, with NULL for the ~~right~~ ^{left} table columns.

select columns
from table 1 AS A
Full outer Join table 2 AS B
ON A.common-column = B

SQL Joins

- Combines tables
- Join 2 or more tables
- It affects the rows
- Full outer join: return all rows from the both tables
- Left Join: returns all rows from left table & only matching rows from the right table
- There must be a common column
- Inner Join: return matching rows.
- Joins give us the ability to combine multiple tables based on a specific
- If no match the NULL (Left & full outer Join)

NULL

- | | | |
|-----------------|---------------|------------------|
| - Blank | - no result | - no data |
| - no value | - empty space | - unknown val |
| - missing value | - void | - no information |

students		sports		sports_name
students-id	name	sports-id	student-id	
1	Sam	10	1	chess
2	Lulu	11	2	Debate
3	Ben	12	4	soccer

select column1,
 column2,
 column3,
 column4

from table1 AS A
 full outer join table2 AS B
 ON A.common-column = B.common-column;

select student-id,
 name,
 sports-id,
 sports-name

from students AS A
 full outer join sports AS B
 ON A.students-id = B.students-id;

Students-id	name	sport-id	sports-name
1	Sam	10	chess
2	Lulu	11	Debate
3	Ben	NULL	NULL
4	NULL	12	soccer